Lecture 2
Model Performance

CSC1104: Data Warehousing and Data Mining
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Prof. Mark Roantree
COMSCI4 @ Dublin City University



Overview
Evaluation Measures
Confusion Matrix
Training and Testing Methods
Comparing Classifiers
Receiver Operating Characteristic Curves
Ensemble Models
Bagging
Boosting
Random Forests

Agenda



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2. **Evaluation Measures**
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4. **Ensemble Models**
   - Bagging
   - Boosting
   - Random Forests

Motivation



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- How can one estimate classifier accuracy when predicting the purchasing behavior of future customers or future (unseen) customer data.
- Many try different methods to build more than one classifier and then need to compare their accuracy.
- But what is accuracy? How can we estimate it?
- Are some measures of a classifier's accuracy more appropriate than others?
- How can we obtain a reliable accuracy estimate?

Training and Testing Datasets



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- Measures include **accuracy** (also known as recognition rate), **sensitivity** (or recall), **specificity**, **precision** and **F-score**.
- While **accuracy** is a specific measure, the word *accuracy* is also a general term for the classifier's predictive abilities.
- Using training data to build a classifier and then estimate accuracy can result in misleading optimistic estimates due to over specialisation of the learning algorithm to the data.
- Instead, measure the classifier accuracy on a test set consisting of class-labeled tuples that were not used to train the model.

Evaluating Classifier Performance

Terminology Recap!

- We refer to *positive* tuples as those of the main class of interest and *negative* tuples for all others.
- Given two classes, the positive tuples could be: *buys computer = yes* while negative tuples are: *buys computer = no*.
- For a classifier on a test set of labeled tuples: P is the number of positive tuples and N is the number of negative tuples.
- For each tuple, compare the classifier class label prediction with the tuple's known class label.



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Evaluation Measures





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Evaluation Building Blocks

- True Positives (TP) refer to the *positive* tuples that were correctly labeled by the classifier.
- True Negatives (TN) are the *negative* tuples that were correctly labeled.
- False positives (FP) are the negative tuples that were *incorrectly* labeled as *positive* (tuples of *buys computer = no* for which the classifier predicted *buys computer = yes*).
- False negatives (FN) are the positive tuples that were mislabeled as negative. In other words, a test for a disease returned negative while the patient was actually positive.



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Measures

Measure	*Formula*
accuracy, recognition rate	$\frac{TP+TN}{P+N}$
error rate, misclassification rate	$\frac{FP+FN}{P+N}$
sensitivity, true positive rate, recall	$\frac{TP}{P}$
specificity, true negative rate	$\frac{TN}{N}$
precision	$\frac{TP}{TP+FP}$
F, F_1, F-score, harmonic mean of precision and recall	$\frac{2 \times precision \times recall}{precision + recall}$

Figure 1: Metrics Using the TP, TN, FP, FN building blocks.



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Confusion Matrix

- A **confusion matrix** is used to analyse how well the classifier recognises tuples of different classes.
- *TP* and *TN* indicate when the classifier is making correct decisions, while *FP* and *FN* indicate that the classifier is mislabeling.
- Given m classes where $m \geq 2$, a confusion matrix is a table of at least size $m \times m$.
- An entry, $CM_{i,j}$ in the first m rows and m columns indicates the number of tuples of class i that were labeled by the classifier as class j.
- Ideally, $i = j$



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Sample Confusion Matrix where m=3

	Class A	Class B	Class C	Total
Class A	45	2	3	50
Class B	12	38	0	50
Class C	3	3	44	50

- Where does the classifier perform well?
- Where does the classifier perform badly?
- Anything interesting? Class B samples?



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Confusion Matrix Accuracy

- For a classifier to have good accuracy, most tuples would be represented along the diagonal ($\searrow$) of the confusion matrix, from $CM_{1,1}$ to $CM_{m,m}$ with the remaining entries being 0 or close to 0.
- Ideally, *FP* and *FN* are close to zero.



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Confusion Matrix Counts

The table may have additional rows or columns to provide totals: *P* and *N* are shown here.

		Predicted class: *yes*	Predicted class: *no*	Total
Actual class	*yes*	*TP*	*FN*	*P*
	no	*FP*	*TN*	*N*
	Total	P'	N'	$P + N$

Figure 2: Confusion matrix for $m = 2$

- P' is the number of tuples that were labeled as positive ($TP + FP$)
- N' is the number of tuples that were labeled as negative ($TN + FN$)
- Total tuple count = $(TP + FN + FP + TN) = (P + N) = (P' + N')$.



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Accuracy

$$accuracy = \frac{TP + TN}{P + N} \quad (1)$$

- Classifier accuracy for a given test set is the percentage of test tuples that are correctly classified
- Also known as the *overall recognition rate*, it reflects the classifier's ability to recognise tuples of all classes.
- A confusion matrix example for the 2 classes *buys computer* $=$ *yes* (positive) and *buys computer* $=$ *no* (negative) is shown in **figure 3**, with totals, recognition rates per class and overall accuracy shown.
- Analysing the confusion matrix, we see that it mislabeled 412 *no* tuples as *yes*.
- Accuracy is most effective when the class distribution is relatively **balanced**.



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Matrix and Accuracy

Confusion matrix for 2 classes *buys computer* $=$ *yes* (positive) and *buys computer* $=$ *no* (negative).

Classes	*buys_computer* $=$ *yes*	*buys_computer* $=$ *no*	*Total*	*Recognition (%)*
buys_computer $=$ *yes*	6954	46	7000	99.34
buys_computer $=$ *no*	412	2588	3000	86.27
Total	7366	2634	10,000	95.42

Figure 3: An entry in row *i*, column *j* shows the number of tuples of class *i* that were labeled as class *j*.



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Error Rate

$$error_rate = \frac{FP + FN}{P + N} \quad (2)$$

- Compute the error rate of a classifier, M, as: $1 - accuracy(M)$, where $accuracy(M)$ is the accuracy of M.
- If we were to use the training set (instead of a test set) to estimate the error rate of a model, this quantity is known as the *resubstitution error*.
- This error estimate is optimistic of the true error rate (and corresponding accuracy) because the model is not tested on any samples that it has not already seen.



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Class Imbalance

- Consider a situation where the main class of interest is rare: the data set distribution reflects a significant majority as *negative* and a minority as *positive*.
- In medical data, there may be a rare class, such as *cancer* $=$ *true*.
- Assume training a classifier for medical data, where the class label attribute is *cancer* and the possible class values are *yes* and *no*.
- An accuracy rate of 97% makes the classifier look great but what if only 3% of the training tuples are actually cancer?
- *What happened here?*
- Measures are needed to determine how well the classifier recognises positive tuples (*cancer* $=$ *yes*) and how well it recognises negative tuples (*cancer* $=$ *no*).



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Sensitivity

$$sensitivity = \frac{TP}{P} \quad (3)$$

- **Sensitivity** is also referred to as the true positive (recognition) rate.
- This represents the *proportion* of positive tuples that are correctly identified.



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Specificity

$$specificity = \frac{TN}{N} \quad (4)$$

- **Specificity** is the true negative rate.
- This represents the *proportion* of negative tuples that are correctly identified.



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Sensitivity and Specificity Example

Classes	*yes*	*no*	*Total*	*Recognition (%)*
yes	**90**	**210**	300	30.00
no	**140**	**9560**	9700	98.56
Total	230	9770	10,000	96.40

Figure 4: Confusion matrix for: *cancer* = *yes* and *cancer* = *no*.

- **Sensitivity** of the classifier is 90/300 = 30.00%.
- **Specificity** is 9560/9700 = 98.56%.
- Overall **accuracy** is 9650/10000 = 96.50%.
- Classifier has high accuracy but poor ability to correctly label the positive (rare) class, with low sensitivity.
- It has high specificity, meaning that it can accurately recognise negative tuples.



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Precision and Recall

- The **precision** and **recall** measures are also widely used in classification.
- Precision can be considered a measure of *exactness*: the percentage of tuples labeled as positive, that are actually positive.
- Calibrate your model so that it picks very few positives? What are the negatives?
- Recall is a measure of *completeness*: the percentage of positive tuples that are labeled as positive.



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Precision

$$precision = \frac{TP}{TP + FP} \quad (5)$$

- The precision of the classifier in **figure 3** for the yes class is 90/230 = 39.13%.



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Recall

$$recall = \frac{TP}{TP + FN} = \frac{TP}{P} \quad (6)$$

- Recall is the same as sensitivity or the true positive rate.
- In **figure 3**, the recall is 90/300 = 30.00%, which is also the same calculation for sensitivity.



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Interpreting *P* and *R*

- A perfect **precision** score of 1.0 for a class *C* means that every tuple labeled as class *C* actually belongs to class *C*.
- *However*, it says nothing about the number of class *C* tuples mislabeled.
- If only 1% of samples were positive, it may be easy to label all samples as positive!
- A perfect **recall** score of 1.0 for *C* means that every item from class *C* was labeled as *C*.
- *However*, it says nothing about the number of other tuples were incorrectly labeled as *C*.



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P vs *R*

- There tends to be an inverse relationship between precision and recall, where it is possible to increase one at the cost of reducing the other.
- Precision and recall scores are typically used together, where precision values are compared for a fixed value of recall, or vice versa.
- For example, we may compare precision values where $recall = 0.75$.



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$F-score$

$$F = \frac{2 \times precision \times recall}{precision + recall} \quad (7)$$

- An alternative way to use precision and recall is to combine them into a single measure: the F measure
- The F measure is the harmonic mean of precision and recall*.
- The equation gives equal weight to precision and recall . . . often used in cases where data is imbalanced.

**Proof is an interesting exercise but outside the scope of this course*



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Random Subsampling

- The **holdout** method is where data are randomly partitioned into two independent sets: a *training* set and a *test* set.
- The training set is used to derive the model but accuracy is measured with the test set: the estimate is *pessimistic* because only a portion of the initial data is used to derive the model.
- **Random subsampling** is a variation of the holdout method in which the holdout method is repeated k times.
- The overall accuracy estimate is taken as the *average* of the accuracies obtained from each iteration.



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Cross-Validation

- In k-fold cross-validation, data are randomly partitioned into k mutually exclusive subsets or *folds* $D_1, D_2, \ldots D_k$, each of (approximately) equal size.
- Training and testing is performed k times.
- In iteration i, partition D_i is reserved as the test set, and the remaining partitions are collectively used to train the model.
- Unlike the holdout and random subsampling methods, each sample here is used the same number of times for training and once for testing.



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Accuracy

- For classification, the accuracy measure is the overall number of correct classifications from the k iterations, divided by the total number of tuples in the initial data.
- Every tuple is tested!



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Comparing Classifiers





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Cost Benefit

- TPs, TNs, FPs, and FNs are used to assess the costs and benefits (or risks and gains) associated with a classification model.
- The cost associated with a false negative (incorrectly predicting a cancerous patient is healthy) is far greater than a false positive (incorrectly labeling a healthy patient as cancerous).
- In these cases, we *outweigh* one type of error over another by assigning a different *cost* to each incorrect classification.
- Similarly, benefits associated with a true positive decision may be different than those of a true negative.



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Receiver Operating Characteristic curves

- ROCs are a useful visual tool for comparing the accuracy of two classification models.
- The ROC curve shows the trade-off between the true positive rate (*TPR*) and the false positive rate (*FPR*).
- *TPR* is the proportion of positive (yes) tuples that are correctly labeled by the model.
- *FPR* is the proportion of negative (no) tuples that are mislabeled as positive.



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Applying Existing Metrics

- We have TP, FP, P, and N as the number of true positive, false positive, positive, and negative tuples, respectively.
- We have $TPR = TP/P$, which is sensitivity.
- We can compute $FPR = FP/N$, which is $1 - specificity$.



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Visualising Accuracy

- For a 2-class problem, an ROC curve visualises the trade-off between: the rate at which the model accurately recognises positive cases against the rate at which it mistakenly identifies negative cases (as positive) for different portions of the test set.
- An increase in *TPR* comes at the cost of increasing in *FPR*.
- For example, you can calibrate your model to have less FNs but this will increase the number of FPs.
- The *area under* the ROC curve is a measure of the accuracy of the model.



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Creating an ROC Plot

- To plot an ROC curve for a given classification model *M*, the model must return a probability of the predicted class for each test tuple.
- For example, patient A is 85% likely to test positive; patient B is 10%, C is 55%, ...
- With this information, rank & sort samples where the sample most likely to belong to the positive class is placed at the top of the list.
- Thus, the tuple that is least likely to belong to the positive class, is placed at the bottom of the list.



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ROC Table

Tuple #	Class	Prob.	TP	FP	TN	FN	TPR	FPR
1	P	0.90	1	0	5	4	0.2	0
2	P	0.80	2	0	5	3	0.4	0
3	N	0.70	2	1	4	3	0.4	0.2
4	P	0.60	3	1	4	2	0.6	0.2
5	P	0.55	4	1	4	1	0.8	0.2
6	N	0.54	4	2	3	1	0.8	0.4
7	N	0.53	4	3	2	1	0.8	0.6
8	N	0.51	4	4	1	1	0.8	0.8
9	P	0.50	5	4	0	1	1.0	0.8
10	N	0.40	5	5	0	0	1.0	1.0

Figure 5: Patients (1-10) sorted by decreasing score, where the score is the value returned by the probabilistic classifier under evaluation.

- Figure 5 shows the **probabilities** (column 3) for 10 patients, returned by a classifier.
- Column 2 is the **actual** class label for the patient: with 5 positive and 5 negative samples: $P = 5$ and $N = 5$



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Table Usage Explained

$TP + FP + TN + FN = 10$ for every row!

Tuple #	Class	Prob.	TP	FP	TN	FN	TPR	FPR
1	P	0.90	1	0	5	4	0.2	0
2	P	0.80	2	0	5	3	0.4	0
3	N	0.70	2	1	4	3	0.4	0.2
4	P	0.60	3	1	4	2	0.6	0.2
5	P	0.55	4	1	4	1	0.8	0.2
6	N	0.54	4	2	3	1	0.8	0.4
7	N	0.53	4	3	2	1	0.8	0.6
8	N	0.51	4	4	1	1	0.8	0.8
9	P	0.50	5	4	0	1	1.0	0.8
10	N	0.40	5	5	0	0	1.0	1.0

Figure 6: Goal is to compute graph points (TPR,FPR)



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Creating an ROC Plot

- Examine the *known class* label of each tuple to determine the values of the remaining columns, *TP*, *FP*, *TN*, *FN*, *TPR*, and *FPR*.
- We start with #1, which has the highest probability score, and use it as the threshold, so $t = 0.9$.
- Thus, the classifier considers tuple 1 to be positive and all the others to be negative.
- Since the actual class label is positive, we have a true positive, so $TP = 1$ and $FP = 0$.
- Of the remaining 9 (classified as negative), 5 are negative, so $TN = 5$; 4 are positive, so $FN = 4$.



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We can therefore compute:

$$TPR = \frac{TP}{P} = \frac{1}{5} = 0.2$$
$$FPR = \frac{FP}{N} = \frac{0}{5} = 0 \quad (8)$$

we have the point (0.2,0) for the ROC curve.



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Th

Step 2

- Next, threshold $t = 0.8$ (probability for tuple 2), so this tuple is now also considered positive, while tuples 3 through 10 are considered negative.
- The actual class label of tuple 2 is positive, thus $TP = 2$.
- When the remaining columns are computed, the result is point (0.4,0).



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Step 3

- Next, $t = 0.7$, the probability value returned by the classifier for that tuple.
- Thus, tuple 3 is considered positive but its actual label is negative, and so it is a false positive.
- Thus, *TP* stays the same and *FP* increments so that $FP = 1$.
- When the remaining columns are computed, the result is point (0.4,0.2).



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ROC Plot

The resulting ROC graph after each of the 10 points is computed, is the broken line shown below.

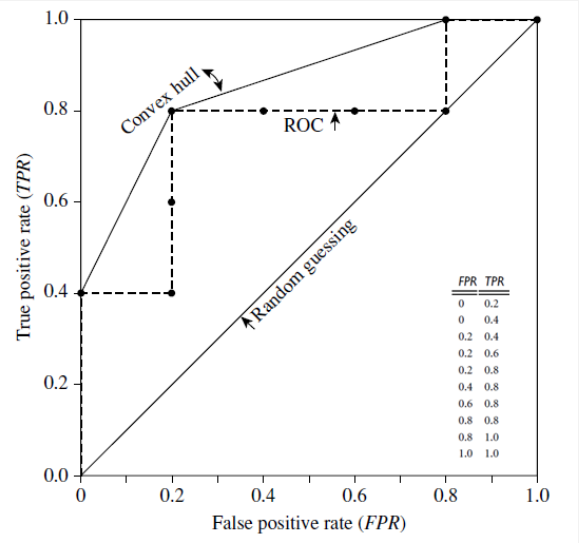


Figure 7: ROC curve for the data in figure 5



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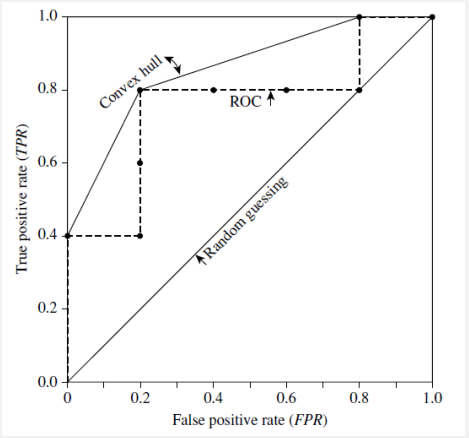


Figure 8: ROC curve for the data in figure 5

- To obtain a curve out of these points, the most common method is to use a convex hull (shown).
- The plot also shows a diagonal line where for every *true* positive, we are just as likely to encounter a *false* positive.
- For comparison, this line represents random guessing.



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Algorithm

The vertical axis of an ROC curve represents *TPR*; the horizontal axis is *FPR*. To plot an ROC curve for M:

- Starting at the bottom left corner ($TPR = FPR = 0$), read the actual class label at the top of the list.
- If we have a true positive, then *TP* and thus *TPR* increase.
- On the graph, **move up** and plot the point.
- If the model classifies a negative tuple as positive, this a false positive, and so both *FP* and *FPR* increase.
- On the graph, we **move right** and plot the point.
- Repeat for each of the test tuples in *ranked* order, each time moving up on the graph for a true positive or toward the right for a false positive.



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ROC Plot Comparison

The diagonal line represents random guessing: the closer the ROC curve of a model is to the diagonal line, the less accurate the model.

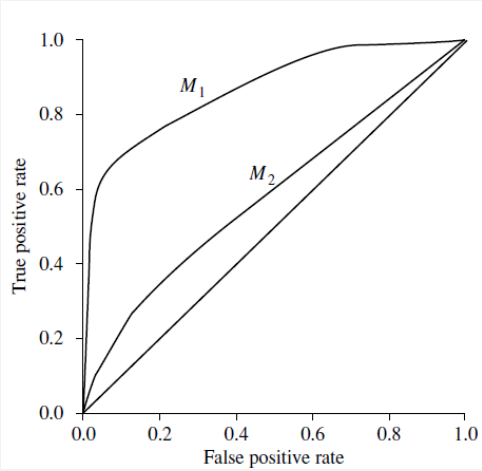


Figure 9: ROC curves of two classification models, M1 and M2. The closer an ROC curve is to the diagonal line, the less accurate the model is. Thus, M1 is more accurate here.



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ROC Plot Analysis

- If the model is really good, we are more likely to encounter true positives as we move down the ranked list: the curve moves steeply up from zero.
- Later, as we start to encounter fewer and fewer true positives, and more and more false positives, the curve eases off and becomes more horizontal.
- To assess the accuracy of a model, we can measure the area under the curve.
- The closer the area is to 0.5, the less accurate the corresponding model is.
- A model with perfect accuracy will have an area of 1.0.



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More Complex Comparisons

To measure the area under the curve, software packages are able to perform such calculation.

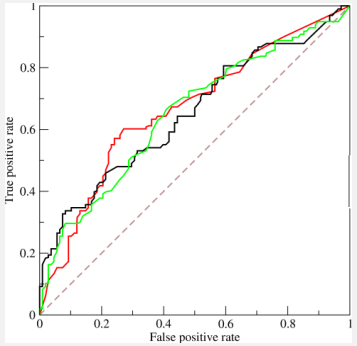


Figure 10: It may not be easy to determine the best model visually.



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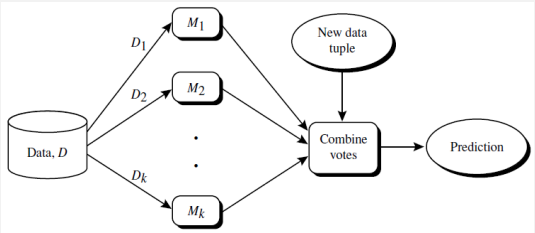
AUC Overview and Usage

- Accuracy is misleading when one class dominates: 99% healthy and 1% rare disease) so predicting healthy every time achieves 99% accuracy.
- AUC evaluates ranking quality regardless of class proportions: AUC works well with imbalanced data!
- An AUC Score of 0.85 means there is an 85% chance the model will rank a randomly chosen positive instance higher than a randomly chosen negative instance.
- Accuracy measures are: *correct* or *incorrect*.
- AUC is generally used in 2-class predictions.



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Ensembles





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Introducing Ensembles Models

- An ensemble **combines** a series of k learned models (base classifiers), $M_1, M_2, \ldots M_k$, to create an improved **composite** classification model, M*.
- A given data set D, is used to create k training sets, $D_1, D_2, \ldots D_k$, where $D_i (1 \leq i \leq k-1)$ is used to generate classifier M_i.
- Given a new data tuple, the base classifiers each *vote* by returning a class prediction.
- The ensemble returns a class prediction based on the *votes* of the base classifiers.



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Ensembles

Bagging, boosting, and random forests are examples of ensemble methods.

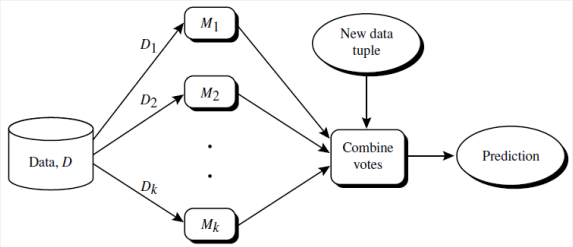


Figure 11: Increasing classifier accuracy: Ensemble methods



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Diversity = Stronger Models

An ensemble tends to be more accurate than its base classifiers.

- Consider an ensemble that performs majority voting:
  - To classify sample X, M* collects predictions returned from base classifiers and outputs the *majority* class.
  - The base classifiers will make mistakes but the ensemble will misclassify X only if $>$ 50% of base classifiers err.
- Ensembles yield better results when there is significant *diversity* among the models.
- Ideally, there is little correlation among classifiers.



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Ensemble Decision Boundary

To help illustrate the power of an ensemble, consider a simple two-class problem described by two attributes, $x1$ and $x2$. The problem has a linear decision boundary.

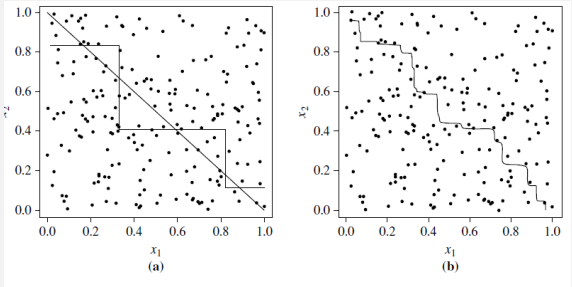


Figure 12: (a) shows the decision boundary of a decision tree classifier on the problem; (b) shows the decision boundary of an ensemble of decision tree classifiers on the same problem.

The ensemble's decision boundary has a finer resolution and is better than any single classifier.



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Bagging

- **Bagging** stands for bootstrap aggregation.
- **Bootstrap** samples training tuples uniformly with replacement: the same tuple is equally likely to be selected again and re-added to the training set.
- **Bagging** works as a method of increasing accuracy where the final diagnosis is made based on a majority.
- Each classifier gets an equal vote.
- Intuitively, a majority vote made by a large group of doctors may be more reliable than a majority vote made by a small group.



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Bagging Algorithm

- For iteration $i (i = 1, 2, \ldots, k)$, a training set D_i, of d tuples is sampled with replacement from the original set D.
- Each training set is a bootstrap sample: some tuples in D may not be included in D_i, whereas others may occur more than once.
- A classifier M_i is learned for each training set D_i.
- To classify an unknown X, each classifier M_i returns its class prediction which counts as one vote.
- The bagged classifier M*, counts the votes and assigns the class with the most votes to X.
- Bagging can be used for the **prediction of continuous values** by computing the average value of each prediction for a given test tuple.



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Boosting

- Unlike Bagging, weights are assigned to the value of **each** classifier's decision, based on the *accuracies of decisions*.
- In **Boosting**, the final diagnosis is a combination of the *weighted* diagnoses.
- Weights are also assigned to each training tuple.



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Iterative Learning

- A series of k classifiers is iteratively learned.
- After a classifier M_i is learned, weights are updated to allow the subsequent classifier M_{i+1} to observe those training tuples *misclassified* by M_i.
- The final boosted classifier M* combines the votes of each individual classifier, where the weight of each classifier's vote is a function of its accuracy.



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Adaptive Boosting

AdaBoost is a popular algorithm to boost the accuracy of a learning method.

- Assume D to be a data set of d class-labeled tuples, $(X_1, y_1), (X_2, y_2), \ldots, (X_d, y_d)$, where y_i is the class label of tuple X_i.
- X is any number of attributes and d is sample count.
- Initially, AdaBoost assigns each training tuple an equal weight of $1/d$.
- Generating k classifiers for the ensemble requires k iterations during the algorithm: each iteration produces a single classifier.



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AdaBoost

- At iteration i, tuples from D are sampled to form a training set D_i of size d.
- Replacement sampling is used: the same tuple may be selected more than once.
- A tuple's chance of selection is based on its weight.
- A classifier M_i is derived from the training tuples of D_i.
- Its error is then calculated using D_i as a test set.
- Training tuple weights are then adjusted according to how they were classified.



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AdaBoost

Goal: build a series of classifiers that complement each other.

- If a tuple was incorrectly classified, its weight is increased.
- If a tuple was correctly classified, its weight is decreased.
- A tuple's weight reflects how difficult it is to classify: the higher the weight, the more often it has been misclassified.
- These weights will be used to generate the training samples for the classifier of the next round.
- The basic idea is that when we build a classifier, we want it to focus more on the misclassified tuples of the previous round.
- Some classifiers may be better at classifying *difficult* tuples.



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Model Performance

To compute the error rate of model M_i, sum the weights of each of the tuples in D_i that M_i misclassified.

$$error(M_i) = \sum_{j=1}^{d} w_j \times err(X_j) \tag{9}$$

- where $err(X_j)$ is the misclassification error of tuple X_j: if misclassified, then $err(X_j) = 1$; otherwise, $err(X_j) = 0$.
- If the performance of classifier M_i is so poor that its error exceeds 0.5, then we abandon it.
- Instead, we try again by generating a new D_i training set, from which we derive a new M_i.



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Tuples Weights

- The error rate of M_i affects how the weights of the training tuples are updated.
- If a tuple at iteration i was correctly classified, its weight w_i is reduced by the following multiplier:

$$w_i \times \frac{error(M_i)}{1 - error(M_i)} \quad (10)$$

- After weights of correctly classified tuples are updated, the weights for *every* tuple are normalised so their sum is unchanged.



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Tuples Weight Normalisation

- Normalise weights: multiply each weight by the sum of the **old weights**, *divided by* the sum of the **new weights**.
- Weights of misclassified tuples are increased; weights of correctly classified tuples are decreased.

$$0.25 \times \frac{0.1}{0.9} \quad (11)$$

Samp.	W1	M1 Prediction	Decrease By	W2	W2(norm)
1	0.25	Correct	0.027777778	0.222222222	0.242424242
2	0.25	Correct	0.027777778	0.222222222	0.242424242
3	0.25	Correct	0.027777778	0.222222222	0.242424242
4	0.25	Incorrect	0	0.25	0.272727273
	1			0.916666667	1

Figure 13: The M1 error rate of 0.1 is used to compute new weights for samples 1-3, before normalisation of all tuples.



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Decision Process

Once boosting is complete, how does the ensemble predict the class label of a new tuple X?

- Unlike bagging, where classifiers were equal, boosting assigns a weight to each classifier's vote, based on the classifier's performance.
- The lower the error rate, the more accurate it is, and thus, the higher its voting weight.
- The weight of classifier M_i's vote is:

$$log\frac{1 - error(M_i)}{error(M_i)} \quad (12)$$

$$log\frac{1 - 0.5}{0.5} = log(1.0) = 0 \quad (13)$$

$$log\frac{1 - 0.8}{0.8} = log(0.25) = negative\ value \quad (14)$$

$$log\frac{1 - 0.2}{0.2} = log(4) = positive\ value \quad (15)$$



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Final Selection

- For each class c_i:
  calculate the *sum of the weights* of each classifier that assigned class c_i to X.
- The class with the highest sum is chosen as the class prediction for X.



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Boosting v Bagging

- Because of the way boosting focuses on the misclassified tuples, it risks overfitting the resulting composite model to the data.
- Thus, the resulting *boosted* model may be less accurate than a *single* model derived from the same data.
- Bagging is less susceptible to model overfitting.
- While both can significantly improve accuracy in comparison to a single model, boosting tends to achieve greater accuracy.



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Random Forests

- In **Random Forests**, each classifier in the ensemble is a decision tree so that the collection of classifiers is a *forest*.
- Individual decision trees are generated using a *random selection* of attributes at each node to determine the split.
- The same sample distribution is given to all trees in the forest.
- During classification, each tree votes and the most popular class is returned.
- Random forests can be built using bagging together with a *random* attribute selection strategy.



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Algorithm (Trees are small)

Assume a training set D, generate k decision trees for the ensemble as follows.

- For iteration $i (i = 1, 2, \ldots, k)$, a training set D_i of d tuples is sampled with replacement from D.
- Reminder: some tuples may occur more than once in D_i, others may be excluded.
- F is the number of attributes used to determine the split at each node, where F is *much smaller* than total attributes.
- To construct a decision tree classifier M_i, for each node, randomly select F attributes as candidates for the split.



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Random Forests v AdaBoost

- Random forests are comparable in accuracy to AdaBoost, yet are more robust to errors and outliers.
- The generalisation error for a forest converges as long as the number of trees in the forest is large: overfitting is not a problem.
- Because random forests consider many fewer attributes for each split, they are efficient on very large datasets: often faster than either bagging or boosting.



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Random Forest Characteristics

- Random forest accuracy depends on the strength of the *individual classifiers* and a measure of the *dependence* between them.
- The goal is to maintain the strength of individual classifiers without increasing their correlation (significant diversity).
- Random forests are insensitive to the number of attributes selected for consideration at each split.
- An interesting empirical observation: using a **single** random input attribute may result in higher accuracy than when using several attributes (tree stumps).
- Feature Importance: random forests give internal estimates of *variable importance*.



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